Walls Have Ears: Wireless-tapping

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Introduction

The purpose of this lab is to inspired interests in mobile security by demonstrating the possibility of wireless tapping. We will explore some security vulnerabilities that are associated with wireless based VoIP services. For example, a third party may be able to intercept a phone conversation and replay it due to the openness of wireless communications.

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1 VoIP Server Setup (for Instructor)

For demonstration and interest inspiration purposes, it is preferred that the instructor (not the students) would setup a VoIP server on a wireless router or a computer (that is connected to the router). If resources and time permit, students can setup the VoIP server. In the following, we will configure a Buffalo WZR-HP-AG300H router/access point (AP) as the VoIP server as an example. The instructor can use other routers if they are compatible with OpenWRT. Alternatively, the VoIP server can be setup on a Linux computer using Docker (`https://hub.docker.com/r/andrius/asterisk`).

1.1 Install Asterisk

All the packages needed our example are located in the following folder: `http://csfaculty.tcu.edu/ma/eureka/?dir=Walls-Have-Ears`. Flash your router with OpenWRT if you have not done so previously. The instructions for flashing a router with OpenWRT can be found in `https://eurekalabs.net/lab/5f306fbf0599dd0208c67100`.

We now proceed to install Asterisk on an OpenWRT enabled router. You need to download Asterisk related packages (in our example, these packages are available at `http://csfaculty.tcu.edu/ma/eureka/?dir=Walls-Have-Ears%2Fvoip-openwrt-packages`). Assume you place all the downloaded *ipk* files under a folder named *packages*. You can upload the packages folder to the */tmp* folder of the router via the following command:

```
scp -r ./packages root@192.168.1.1:/tmp
```

After uploading the ipk files on to your router, you can start installing them with *opkg*. On your router, type the following command:

```
opkg install *.ipk
```

If you receive no errors, Asterisk has been successfully installed on your router.

1.2 Configure Asterisk

There are two files need to be configured for Asterisk: *extensions.conf* and *sip.conf*. Example of these two files can be seen at `http://csfaculty.tcu.edu/ma/eureka/?dir=Walls-Have-Ears%2Fexample-conf`.

Go to the */etc/asterisk* directory on the router. Edit the *sip.conf* based on the following example (you may want to back up sip.conf before making changes).

```
[general]
tos_sip=cs3 ; set sip traffic tos bit
tos_audio=ef ; set sip audio dscp bit to ef
bindaddr=192.168.1.1
bindport=5060

[20]        ; (20 is the phone number for John Doe)
type=friend ; can place and receive calls
context=peers ; where to place the call
host=dynamic ; can connect from any ip address
secret=eureka ; password
callerid=``John Doe'' (You can name the caller ID as you prefer)
```

```
[21]
type=friend
context=peers
host=dynamic
secret=labs
callerid=``Jane Doe''   ; (21 is the phone number for Jane Doe)

[22]
type=friend
context=peers
host=dynamic
secret=iOS
callerid=``iPad User''  ; (22 is the phone number for iPad User)
```

Edit extensions.conf based on the following example.

```
[peers]
exten => 20,1,Dial(SIP/20,10)
exten => 20,n,Hangup()
exten => 21,1,Dial(SIP/21,10)
exten => 21,n,Hangup()
exten => 22,1,Dial(SIP/22,10)
exten => 22,n,Hangup()
[incoming] include =>peers
```

Start Asterisk in verbose debugging mode as follows,

```
asterisk -cv
```

The option *v* means Asterisk will be verbose in its display. When you issue this command, a great deal of information will scroll past quite quickly. Eventually, you will receive a prompt stating CLI. Do not quit the Asterisk terminal during the entire lab session. Reference to Asterisk CLI commands: `http://www.automated.it/asterisk/lah-3-6-05_3.html`.

1.3 Setup a Wireless Network

Next, you will need to setup a wireless network to VoIP clients to connect to the VoIP server. For demonstration purposes, the wireless network does not need to enforce security mechanisms such as WEP or WPA. The related commands (using *uci*) of setting up a wireless network can be found in `https://eurekalabs.net/lab/5f306fbf0599dd0208c67100`. You can also use the OpenWRT GUI interface to do so. VoIP clients will connect to your wireless network later. The reason we use non-security is because we want to focus on the VoIP traffic capture in this lab.

1.4 Install and Configure VoIP Client

To make VoIP phone calls, we need to install a softphone program on a smart device or a wireless enabled computer. For example, Bria (previously known as X-Lite, `https://www.counterpath.com/x-lite/`) is a popular app. Its iOS version is available at `https://apps.apple.com/app/id1074064248`, and Android version is available at `https://play.google.com/store/apps/details?id=com.cpc.briax`.

After installation, start the Bria/X-Lite softphone program. You will be prompted to add a SIP account (an example shown in Fig. 1). Supply the required information accordingly. **The account information needs to match what you have in the sip.copf file**. For example, if you choose to use number *21*, you need to use the password as *labs*. You could use a customized display name.

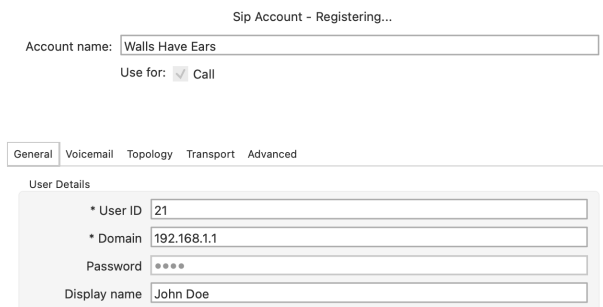

Figure 1: Softphone account configuration.

Connect your softphone device to the wireless network. After starting the X-Lite for a few seconds with the right account information, you should see messages displayed on your Asterisk CLI about a SIP account registration. A complete registration procedure on the router and a desktop client can be seen in Fig. 2.

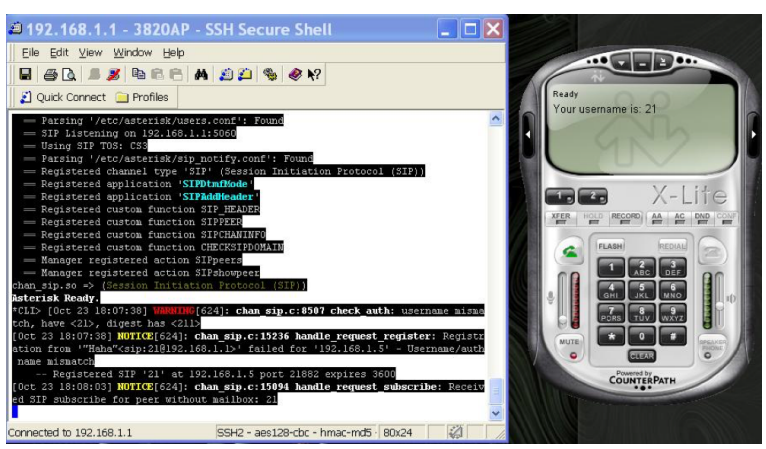

Figure 2: Registration information on the router and a client.

On a successful registration, you should see a message on the console on your router along the lines of “Registered SIP ‘21’ at XXX.XXX.XXX.XXX port”, Where XXX.XXX.XXX.XXX is the IP address of your softphone device. Now, you are ready to make VoIP phone calls. Make a test phone call between your devices before proceeding to next section.

2 “Wireless-tapping”: Walls Have Ears

Since the VoIP data traffics are sent over the air, your phone conversation can be captured and replayed by a third party. In this section, we will capture an on-going VoIP conversation and replay this conversation using Wireshark.

2.1 Eavesdrop on Wireless VoIP Calls

The instructor and a student can initiate a phone conversation using any of the two numbers on two devices assuming that the two devices are ready (i.e., registered with the Asterisk on router/computer).

On a third computer, drive its wireless NIC into the monitor mode (e.g., using *airmon-ng*). Open Wireshark and start capturing wireless data using the interface for about 15 seconds (or you can use airodump-ng to capture the data). Stop the phone conversation and capture. Save the captured data to a pcap file (an example file can be downloaded from `http://csfaculty.tcu.edu/ma/eureka/?dir=Walls-Have-Ears%2Fcaptured-call`). Open the pcap file in Wireshark, go to –> **Telephony** –> **VoIP Calls**. The Wireshark will display the captured phone conversation (an example shown in Fig. 3).

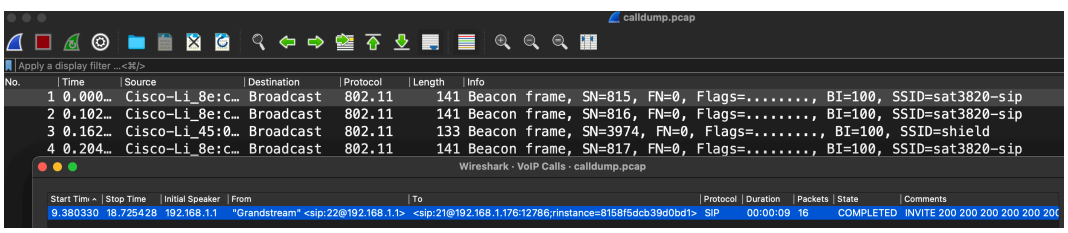

Figure 3: VoIP call shown in Wireshark.

Click the desired phone conversation, select **Playe Streams**. A new window will pop out, which looks Fig. 4. Now, you choose communications (with the wave forms) and play the conversation. Verify your results with the instructor.

2.1.1 Provide a screenshot your captured phone conversation in Wireshark.

2.1.2 What would you do to secure your wireless VoIP conversations?

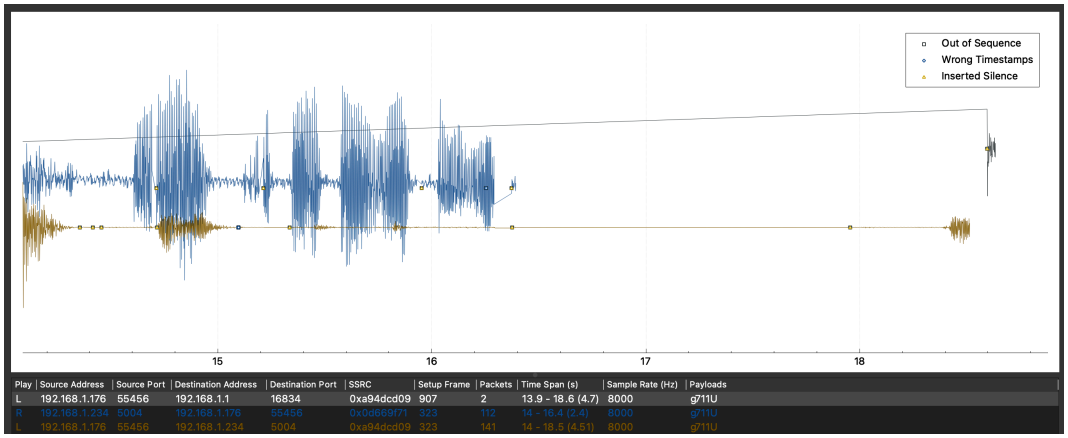

Figure 4: Sound stream of the captured VoIP call.

You could use or WEP or WPA to encrypt your VoIP conversation. Yet, this may degrade the router's performance due to the large amount of computation caused by encryption/decryption.

2.2 VoIP Call Disruption

Make another phone conversation between two devices. Use a third computer to launch a de-authentication attack on the one of the two communicating parties during their conversation.

2.2.1 Describe what happened and how did it happen.

2.2.2 What would you do to prevent such attacks?

3 General Questions

3.0.1 How long did it take you to complete this lab?

3.0.2 Are there any corrections needed to apply to this lab